

1.1 Quadratics

What is the central idea in Quadratics?

Completing the square reveals turning point and range; discriminant b^2-4ac classifies real roots.

9709-01-C1

QUESTION + ANSWER

1.2 Functions

Which mechanism, condition or distinction matters in Functions?

A composite fg means $f(g(x))$ under common notation, with domain restricted so $g(x)$ lies in f 's domain.

9709-02-C2

QUESTION + ANSWER

1.3 Coordinate geometry

Which mechanism, condition or distinction matters in Coordinate geometry?

A circle $(x-a)^2+(y-b)^2=r^2$ has centre (a,b) . A tangent is perpendicular to radius at contact.

9709-03-C2

QUESTION + ANSWER

1.4 Circular measure

A sector has radius 5 cm and angle 1.2 rad. Find arc length and area.

$s=r\theta=6$ cm; $A=\frac{1}{2}r^2\theta=15$ cm².

9709-04-P

QUESTION + ANSWER

1.5 Trigonometry

What is the central idea in Trigonometry?

Use exact unit-circle values and signs by quadrant. Identities $\sin^2x+\cos^2x=1$ and $\tan x=\sin x/\cos x$ simplify expressions.

9709-05-C1

QUESTION + ANSWER

1.1 Quadratics

Which mechanism, condition or distinction matters in Quadratics?

A quadratic inequality changes sign at roots when they exist; use a sketch or interval sign test, including endpoints as the question requires.

9709-01-C2

QUESTION + ANSWER

1.2 Functions

For $f(x)=3x-2$, find $f^{-1}(x)$ and $f^{-1}(10)$.

$y=3x-2$ gives $x=(y+2)/3$; $f^{-1}(10)=4$.

9709-02-P

QUESTION + ANSWER

1.3 Coordinate geometry

What is the central idea in Coordinate geometry?

For a line gradient $m=(y_2-y_1)/(x_2-x_1)$; parallel lines share gradient and perpendicular non-vertical gradients multiply to -1 .

9709-03-C1

QUESTION + ANSWER

1.4 Circular measure

Which mechanism, condition or distinction matters in Circular measure?

Convert θ° to $\theta\pi/180$; for a segment subtract triangle area $\frac{1}{2}r^2\sin\theta$ from sector area.

9709-04-C2

QUESTION + ANSWER

1.5 Trigonometry

Solve $\cos x=-1/2$ for $0\leq x<2\pi$.

Reference angle $\pi/3$; cosine negative in quadrants II and III; $x=2\pi/3, 4\pi/3$.

9709-05-P

QUESTION + ANSWER

1.6 Series

What is the central idea in Series?

Arithmetic sequences have $u_n=a+(n-1)d$ and $S_n=n[2a+(n-1)d]/2$.

9709-06-C1

QUESTION + ANSWER

1.1 Quadratics

Solve $x^2-5x+6<0$.

Factor $(x-2)(x-3)<0$; upward parabola negative between roots; $2<x<3$.

9709-01-P

QUESTION + ANSWER

1.2 Functions

What is the central idea in Functions?

A function assigns each allowed input one output. State domain before finding range or inverse; inverse exists on a one-to-one restricted domain.

9709-02-C1

QUESTION + ANSWER

1.3 Coordinate geometry

Find equation of tangent to circle $x^2+y^2=25$ at $(3,4)$.

Radius gradient $4/3$, tangent gradient $-3/4$; $y-4=-3(x-3)/4$.

9709-03-P

QUESTION + ANSWER

1.4 Circular measure

What is the central idea in Circular measure?

In radians, arc length $s=r\theta$ and sector area $A=\frac{1}{2}r^2\theta$. These formulas require θ in radians, not degrees.

9709-04-C1

QUESTION + ANSWER

1.5 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

For trig equations over an interval, find all solutions using symmetry and periodicity; do not accept inverse-calculator principal value alone.

9709-05-C2

QUESTION + ANSWER

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1.6 Series

Which mechanism, condition or distinction matters in Series?

A geometric sequence has $u_n = ar^{(n-1)}$, $S_n = a(1-r^n)/(1-r)$ if $r \neq 1$, and infinite sum $a/(1-r)$ only for $|r| < 1$.

9709-06-C2

[QUESTION + ANSWER](#)

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1.7 Differentiation

Differentiate $y = (2x+1)^3$ and find gradient at $x=1$.

Chain rule $y' = 3(2x+1)^2 \times 2 = 6(2x+1)^2$; at $x=1$ gradient 54.

9709-07-P

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.8 Integration

Which mechanism, condition or distinction matters in Integration?

A substitution reverses the chain rule and usually changes integration limits in a definite integral.

9709-08-C2

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.1 Algebra

Expand $(1+x)^{-2}$ to terms in x^2 and state validity.

Use binomial coefficients: $1 - 2x + 3x^2$; $|x| < 1$.

9709-09-P

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

Which mechanism, condition or distinction matters in Logarithmic and exponential functions?

Exponential growth/decay solves $dy/dt = ky$ and requires the initial condition to determine the multiplier.

9709-10-C2

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.6 Series

Find first six terms' sum for geometric series 2, 6, 18, ...

$a=2$, $r=3$; $S_6 = 2(3^6 - 1)/(3 - 1) = 728$.

9709-06-P

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.7 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

Stationary points satisfy derivative zero, but classify with second derivative or sign change. Related rates require units and sign.

9709-07-C2

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.8 Integration

Evaluate $\int$ from 1 to 3 of $2x \, dx$.

Antiderivative x^2 ; x^2 evaluated 1 to 3 $= 9 - 1 = 8$.

9709-08-P

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.1 Algebra

Which mechanism, condition or distinction matters in Algebra?

Partial fractions simplify rational integration; first factor denominator and match numerator degrees. Factor theorem provides roots and polynomial division reduces degree.

9709-09-C2

[QUESTION + ANSWER](#)

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2.2 Logarithmic and exponential functions

Solve $\ln(x-1) = 2$ for real x .

Domain $x > 1$; exponentiate: $x-1 = e^2$; $x = 1 + e^2$.

9709-10-P

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.7 Differentiation

What is the central idea in Differentiation?

Derivative is instantaneous gradient and rate of change. Power rule $d(x^n)/dx = nx^{n-1}$; use product, quotient and chain rules where needed.

9709-07-C1

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
1.8 Integration

What is the central idea in Integration?

Integration reverses differentiation up to a constant C . For a definite integral use antiderivative limits; signed area below x -axis is negative.

9709-08-C1

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.1 Algebra

What is the central idea in Algebra?

Binomial expansion for non-integer powers is infinite and valid only for $|x| < 1$ in $(1+x)^n$.

9709-09-C1

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

What is the central idea in Logarithmic and exponential functions?

$\ln$ and $\exp$ are inverses on appropriate domains. Log rules combine products into sums and powers into coefficients; do not split $\ln(a+b)$.

9709-10-C1

[QUESTION + ANSWER](#)

FagusED · A-Level 9709
2.3 Trigonometry

What is the central idea in Trigonometry?

At this level, double-angle identities $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = 2 \cos^2 x - 1$ help solve equations and integrate.

9709-11-C1

[QUESTION + ANSWER](#)

2.3 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

State the full interval after algebra and test solutions excluded by any division by $\sin x$ or $\cos x$.

9709-11-C2

QUESTION + ANSWER

2.4 Differentiation

For $x=t^2+1$, $y=t^3$, calculate dy/dx at $t=2$.

$dy/dt=3t^2$, $dx/dt=2t$; $dy/dx=3t/2$; at 2 gradient 3.

9709-12-P

QUESTION + ANSWER

2.5 Integration

Evaluate $\int$ from 0 to 1 of $2x(x^2+1)$ dx.

Substitute $u=x^2+1$, $du=2x$ dx; integrate u from 1 to 2; $[u^2/2]_1^2=3/2$.

9709-13-P

QUESTION + ANSWER

2.6 Numerical solution of equations

Which mechanism, condition or distinction matters in Numerical solution of equations?

Iteration $x_{n+1}=g(x_n)$ is useful if it converges; convergence near a fixed point is favoured when $abs(g'(x))<1$. Report a final answer only after checking required accuracy.

9709-14-C2

QUESTION + ANSWER

3.2 Logarithmic and exponential functions

What is the central idea in Logarithmic and exponential functions?

Use logarithms to linearise exponential equations and transform data when a model $y=Ae^{(kx)}$ is justified.

9709-16-C1

QUESTION + ANSWER

2.3 Trigonometry

Solve $\sin 2x=0$ for $0\leq x<2\pi$.

$2x=n\pi$; $x=n\pi/2$; $x=0,\pi/2,\pi,3\pi/2$.

9709-11-P

QUESTION + ANSWER

2.4 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

Set $dy/dx=0$ carefully because a zero numerator with zero denominator may need separate analysis.

9709-12-C2

QUESTION + ANSWER

2.5 Integration

Which mechanism, condition or distinction matters in Integration?

Area of a region between graphs is integral of upper minus lower curve; split at intersections. For volumes of revolution about an axis use $\pi\int y^2 dx$ when permitted by syllabus.

9709-13-C2

QUESTION + ANSWER

2.6 Numerical solution of equations

Show $f(x)=x^3-2$ has a root between 1 and 2 and perform one Newton step from $x_0=1.5$.

$f(1)=-1$, $f(2)=6$ so a root lies between; $f'=3x^2$;
 $x_1=1.5-(1.5^3-2)/(3\times 1.5^2)\approx 1.2963$.

9709-14-P

QUESTION + ANSWER

3.1 Algebra

Decompose $3/[x(x+3)]$ into simple fractions.

Set $A/x+B/(x+3)$: $3=A(x+3)+Bx$; $x=0$ gives $A=1$, $x=-3$ gives $B=-1$.

9709-15-P

QUESTION + ANSWER

2.4 Differentiation

What is the central idea in Differentiation?

Implicit differentiation differentiates y as $y(x)$, introducing dy/dx . Parametric curves use $dy/dx=(dy/dt)/(dx/dt)$; chain and product rules handle composite products.

9709-12-C1

QUESTION + ANSWER

2.5 Integration

What is the central idea in Integration?

Integration by substitution identifies an inner expression whose derivative also occurs.

9709-13-C1

QUESTION + ANSWER

2.6 Numerical solution of equations

What is the central idea in Numerical solution of equations?

A sign change $f(a)f(b)<0$ for continuous f brackets a root, subject to continuity.

9709-14-C1

QUESTION + ANSWER

3.1 Algebra

What is the central idea in Algebra?

Advanced algebra includes factorisation, partial fractions and binomial series as appropriate.

9709-15-C1

QUESTION + ANSWER

3.1 Algebra

Which mechanism, condition or distinction matters in Algebra?

For a rational expression, divide first if numerator degree is not lower than denominator, then decompose using factors and repeated-factor terms. Validate by recombining or testing convenient x .

9709-15-C2

QUESTION + ANSWER

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3.2 Logarithmic and exponential functions

Which mechanism, condition or distinction matters in Logarithmic and exponential functions?

Differentiation $d(e^{kx})/dx = ke^{kx}$; $d(\ln(ax))/dx = 1/x$ for $x \neq 0$. Check original domain after solving logarithmic expressions.

9709-16-C2

QUESTION + ANSWER

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3.3 Trigonometry

Write $5\cos x + 12\sin x$ as $R\cos(x - \alpha)$ with $R > 0$ and α in first quadrant.

$R = \sqrt{25 + 144} = 13$; $R\cos\alpha = 5$, $R\sin\alpha = 12$; expression $13\cos(x - \arctan(12/5))$.

9709-17-P

QUESTION + ANSWER

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3.4 Differentiation

Differentiate $y = xe^{2x}$ and find y' at $x=0$.

Product and chain:
 $y' = e^{2x} + 2xe^{2x} = e^{2x}(1 + 2x)$; at $x=0$, 1.

9709-18-P

QUESTION + ANSWER

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3.5 Integration

Evaluate $\int$ from 0 to 1 of $1/(x+1)$ dx.

Antiderivative $\ln(ax+1)$;
 $\ln 2 - \ln 1 = \ln 2$.

9709-19-P

QUESTION + ANSWER

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3.6 Numerical solution of equations

Which mechanism, condition or distinction matters in Numerical solution of equations?

Interval bisection gives a guaranteed bracket under continuity and sign change. Numerical answers require a reason for chosen rounding and a check at boundaries.

9709-20-C2

QUESTION + ANSWER

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3.2 Logarithmic and exponential functions

Solve $e^{2x} = 7$ and give exact x.

Take logs: $2x = \ln 7$; $x = (\ln 7)/2$.

9709-16-P

QUESTION + ANSWER

FagusED · A-Level 9709
3.3 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

For equations, list every solution over the specified interval and reject extraneous values from squaring.

9709-17-C2

QUESTION + ANSWER

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3.4 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

Implicit/parametric second derivatives require care: $d^2y/dx^2 = (d/dt(dy/dx))/(dx/dt)$. Identify stationary points and inflection using derivatives and sign tests.

9709-18-C2

QUESTION + ANSWER

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3.5 Integration

Which mechanism, condition or distinction matters in Integration?

For integrals with a linear inner function multiply by reciprocal of its derivative. Improper integrals need limiting processes when bounds or integrand are unbounded.

9709-19-C2

QUESTION + ANSWER

FagusED · A-Level 9709
3.6 Numerical solution of equations

For $f(x) = x^2 - 3$ and $x_0 = 2$, calculate one Newton step.

$f(2) = 1$, $f'(2) = 4$; $x_1 = 2 - 1/4 = 1.75$.

9709-20-P

QUESTION + ANSWER

FagusED · A-Level 9709
3.3 Trigonometry

What is the central idea in Trigonometry?

Use addition formulae, double-angle identities and $R\cos(x - \alpha)$ transformations. A transformed form reveals amplitude and phase; choose α using both sine and cosine coefficients.

9709-17-C1

QUESTION + ANSWER

FagusED · A-Level 9709
3.4 Differentiation

What is the central idea in Differentiation?

Differentiate composite trig, exponential and logarithmic functions with chain, product and quotient rules.

9709-18-C1

QUESTION + ANSWER

FagusED · A-Level 9709
3.5 Integration

What is the central idea in Integration?

Advanced methods include partial fractions, substitution and parts, and trig/exponential integrals.

9709-19-C1

QUESTION + ANSWER

FagusED · A-Level 9709
3.6 Numerical solution of equations

What is the central idea in Numerical solution of equations?

Newton-Raphson updates $x_{n+1} = x_n - f(x_n)/f'(x_n)$; if f' is small it may jump far or fail.

9709-20-C1

QUESTION + ANSWER

FagusED · A-Level 9709
3.7 Vectors

What is the central idea in Vectors?

A line in 3D has $r = a + \lambda b$. Dot product $a \cdot b = |a||b|\cos\theta$ gives angle and perpendicularity.

9709-21-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Vectors?

To intersect two lines, solve all three component equations for their separate parameters; a pairwise apparent intersection may be skew if the third fails.

9709-21-C2

QUESTION + ANSWER

What is the central idea in Complex numbers?

Write $z=x+iy$; modulus and argument give polar form. Multiplication adds arguments and multiplies moduli; conjugate multiplication rationalises division.

9709-23-C1

QUESTION + ANSWER

What is the central idea in Forces and equilibrium?

Draw a free-body diagram and resolve each force into horizontal and vertical components. Equilibrium requires zero resultant in both directions.

9709-24-C1

QUESTION + ANSWER

What is the central idea in Kinematics of motion in a straight line?

Velocity is derivative of displacement; acceleration is derivative of velocity. Integrate acceleration to velocity and velocity to displacement with constants fixed by conditions.

9709-25-C1

QUESTION + ANSWER

What is the central idea in Momentum?

Linear momentum $p=mv$ has direction; impulse $F\Delta t$ changes momentum. For collisions, total momentum is conserved if external impulse is negligible.

9709-26-C1

QUESTION + ANSWER

For $a=(1,2,2)$ and $b=(2,0,0)$, find $\cos\theta$ between them.

$a\cdot b=2$, $abs(a)=3$ and $abs(b)=2$;
 $\cos\theta=2/6=1/3$.

9709-21-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Differential equations?

Model assumptions determine meaningful domain; initial conditions select a particular solution. Exponential change arises when rate is proportional to current amount.

9709-22-C2

QUESTION + ANSWER

Express $(3+4i)/(1-i)$ in $a+bi$ form.

Multiply top and bottom by $(1+i)$;
numerator $(3+4i)(1+i)=-1+7i$; divide 2: $-1/2+(7/2)i$.

9709-23-P

QUESTION + ANSWER

A block weighs 20 N on horizontal surface and is pulled horizontally by 6 N while at rest. State friction and reaction, assuming no other vertical forces.

Friction =6 N opposite pull; normal reaction =20 N.

9709-24-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Kinematics of motion in a straight line?

Constant-acceleration equations require truly constant a ; sign conventions must be maintained in multi-stage travel.

9709-25-C2

QUESTION + ANSWER

What is the central idea in Differential equations?

Separable differential equations rearrange to integrate functions of y and x on opposite sides.

9709-22-C1

QUESTION + ANSWER

Solve $dy/dx=-0.5y$ with $y(0)=8$.

Separate $dy/y=-0.5 dx$;
 $\ln abs(y)=-0.5x+C$; $y=8e^{(-0.5x)}$.

9709-22-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Complex numbers?

Real-coefficient polynomial equations have complex roots in conjugate pairs, a fact useful when solving quadratics and higher-degree equations.

9709-23-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Forces and equilibrium?

Friction is bounded by μR for limiting friction; otherwise its actual magnitude is determined by equilibrium/dynamics, not automatically μR .

9709-24-C2

QUESTION + ANSWER

A particle has $v=4t-2 \text{ m s}^{-1}$ and $s(0)=3 \text{ m}$. Find $s(t)$ and displacement from $t=0$ to 2.

Integrate: $s=2t^2-2t+C$, $C=3$;
 $s(2)-s(0)=4 \text{ m}$.

9709-25-P

QUESTION + ANSWER

4.3 Momentum

Which mechanism, condition or distinction matters in Momentum?

Kinetic energy may decrease even when momentum is conserved, especially when bodies stick.

9709-26-C2

QUESTION + ANSWER

4.3 Momentum

Two 0.5 kg carts, one at 6 m s^{-1} and one stationary, stick. Find speed and lost kinetic energy.

Momentum gives $v = (0.5 \times 6) / 1 = 3 \text{ m s}^{-1}$; KE initial 9 J, final 4.5 J, loss 4.5 J.

9709-26-P

QUESTION + ANSWER

4.4 Newton's laws of motion

What is the central idea in Newton's laws of motion?

Resultant force equals mass times acceleration for a constant-mass body.

9709-27-C1

QUESTION + ANSWER

4.4 Newton's laws of motion

Which mechanism, condition or distinction matters in Newton's laws of motion?

In connected-particle models, write an equation for each body and use a common acceleration; tension is an internal interaction and cancels when treating both as one system. Resistive forces oppose motion, not necessarily velocity axes.

9709-27-C2

QUESTION + ANSWER

4.5 Energy, work and power

What is the central idea in Energy, work and power?

Work by a force is component along displacement times distance. Net work changes kinetic energy; gravitational potential and elastic stores can simplify a mechanics problem when losses are absent.

9709-28-C1

QUESTION + ANSWER

4.4 Newton's laws of motion

A 4 kg mass hangs against a 1 kg mass over a smooth pulley, $g = 10$. Find acceleration.

Driving force difference $(4 - 1)g = 30 \text{ N}$; total mass 5 kg; $a = 6 \text{ m s}^{-2}$ toward 4 kg side.

9709-27-P

QUESTION + ANSWER

5.1 Representation of data

What is the central idea in Representation of data?

Choose displays suited to data type; histogram area, not height, represents frequency if widths differ.

9709-29-C1

QUESTION + ANSWER

4.5 Energy, work and power

A 2 kg object falls 5 m from rest without resistance, $g = 10$. Determine speed using energy conservation.

Loss in GPE $= mg\Delta h = 100 \text{ J}$;
 $\frac{1}{2}mv^2 = 100$; $v = 10 \text{ m s}^{-1}$.

9709-28-P

QUESTION + ANSWER

4.5 Energy, work and power

Which mechanism, condition or distinction matters in Energy, work and power?

Power is work per second or instantaneous Fv when force acts along velocity.

9709-28-C2

QUESTION + ANSWER

5.2 Permutations and combinations

What is the central idea in Permutations and combinations?

Permutations count ordered selections $nPr = n! / (n - r)!$; combinations count unordered groups $nCr = n! / [r!(n - r)!]$.

9709-30-C1

QUESTION + ANSWER

5.1 Representation of data

Two histogram classes have frequencies 12 and 18 with widths 3 and 6. Find both frequency densities.

$12/3 = 4$; $18/6 = 3$.

9709-29-P

QUESTION + ANSWER

5.1 Representation of data

Which mechanism, condition or distinction matters in Representation of data?

Cumulative frequency estimates median and percentiles; mean of grouped data uses midpoints and is approximate. Standard deviation measures spread about the mean, unlike range which relies on extremes.

9709-29-C2

QUESTION + ANSWER

5.2 Permutations and combinations

How many different arrangements of letters LEVEL are there?

Five positions with two L and two E; $5! / (2!2!) = 30$.

9709-30-P

QUESTION + ANSWER

5.2 Permutations and combinations

Which mechanism, condition or distinction matters in Permutations and combinations?

For arrangements with repeated identical objects divide by repeated factorials. Restrictions such as adjacency are often handled by blocks or complement.

9709-30-C2

QUESTION + ANSWER

5.3 Probability

What is the central idea in Probability?

Conditional probability $P(A \text{ given } B) = P(A \cap B) / P(B)$. Independent events satisfy $P(A \cap B) = P(A)P(B)$, which differs from mutually exclusive nonempty events.

9709-31-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Probability?

Tree branches must use updated probabilities after sampling without replacement; check totals sum to one.

9709-31-C2

QUESTION + ANSWER

A bag has three blue and two green counters; draw two without replacement. Find probability of one of each.

Blue-green: $(3/5)(2/4)=3/10$;
green-blue: $(2/5)(3/4)=3/10$; total $3/5$.

9709-31-P

QUESTION + ANSWER

What is the central idea in Discrete random variables?

A probability distribution has nonnegative probabilities summing to 1. Expected value is $\sum xP(X=x)$; variance is $\sum x^2P(X=x) - [E(X)]^2$.

9709-32-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Discrete random variables?

A binomial model requires fixed number of independent trials with constant success probability and two outcomes per trial.

9709-32-C2

QUESTION + ANSWER

What is the central idea in The normal distribution?

Normal distributions are symmetric with mean μ and standard deviation σ ; standardise $z=(x-\mu)/\sigma$ before using tables.

9709-33-C1

QUESTION + ANSWER

For $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$, find $E(X)$ and $\text{Var}(X)$.

$E=0+0.5+0.6=1.1$;
 $E(X^2)=0.5+1.2=1.7$;
 $\text{Var}=1.7-1.1^2=0.49$.

9709-32-P

QUESTION + ANSWER

What is the central idea in The Poisson distribution?

Poisson models counts of events in a fixed interval under independent, constant-rate assumptions.
 $P(X=r)=e^{-\lambda} \lambda^r/r!$, with mean and variance λ .

9709-34-C1

QUESTION + ANSWER

For $X \sim N(50, 10^2)$, express $P(40 < X < 60)$ in standard normal form and estimate it.

z bounds -1 and $+1$; probability $\Phi(1) - \Phi(-1) \approx 0.6827$.

9709-33-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in The normal distribution?

A continuous distribution has $P(X=a)=0$. The normal approximation to a discrete model needs suitable conditions and continuity correction where applicable.

9709-33-C2

QUESTION + ANSWER

What is the central idea in Linear combinations of random variables?

Linearity gives $E(aX+bY)=aE(X)+bE(Y)$. For independent X, Y , $\text{Var}(aX+bY)=a^2\text{Var}(X)+b^2\text{Var}(Y)$; covariance changes this if dependence exists.

9709-35-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in The Poisson distribution?

For twice the exposure, double λ ; a binomial approximation may apply for large n and small p with $\lambda=np$.

9709-34-C2

QUESTION + ANSWER

Calls arrive at mean rate 3 per hour. Find $P(\text{no calls in 20 minutes})$.

Mean for $1/3$ h is $\lambda=1$; $P(X=0)=e^{-1} 1^0/0! \approx 0.3679$.

9709-34-P

QUESTION + ANSWER

What is the central idea in Continuous random variables?

For continuous density f , probability between bounds is area under curve. Normalisation $\int f=1$ fixes constants; $E(X)=\int xf(x)dx$ and $\text{Var}(X)=\int x^2f(x)dx - E(X)^2$.

9709-36-C1

QUESTION + ANSWER

Independent X, Y have $E(X)=3$, $E(Y)=5$, $\text{Var}(X)=2$, $\text{Var}(Y)=4$. Find $E(2X-Y)$ and $\text{Var}(2X-Y)$.

$E=2(3)-5=1$; $\text{Var}=4(2)+4=12$.

9709-35-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Linear combinations of random variables?

Variances are not subtracted when finding difference of independent variables.

9709-35-C2

QUESTION + ANSWER

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6.3 Continuous random variables

Which mechanism, condition or distinction matters in Continuous random variables?

A cumulative distribution $F(x)$ integrates f and is nondecreasing from 0 to 1.

9709-36-C2

QUESTION + ANSWER

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6.4 Sampling and estimation

A normal population has known $\sigma=12$; a random sample $n=36$ has mean 50. Form approximate 95% z interval for μ .

$SE=12/\sqrt{36}=2$;
margin= $1.96 \times 2=3.92$; interval
(46.08,53.92).

9709-37-P

QUESTION + ANSWER

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6.3 Continuous random variables

For $f(x)=kx^2$ on $0 \leq x \leq 1$, find k and $P(X>0.5)$.

Normalise $k \int_0^1 x^2 dx = k/3 = 1$, so $k=3$;
 $P(X>0.5)=1-(0.5)^3=0.875$.

9709-36-P

QUESTION + ANSWER

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6.4 Sampling and estimation

Which mechanism, condition or distinction matters in Sampling and estimation?

A confidence interval combines estimate with critical value times standard error; distinguish confidence of method from probability of a fixed parameter.

9709-37-C2

QUESTION + ANSWER

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6.5 Hypothesis tests

Which mechanism, condition or distinction matters in Hypothesis tests?

Compare p-value with significance level or statistic with critical region; final conclusion must be in context and must not claim H_0 proven. A one-sided test requires a justified directional claim made in advance.

9709-38-C2

QUESTION + ANSWER

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6.4 Sampling and estimation

What is the central idea in Sampling and estimation?

A statistic varies from sample to sample; its sampling distribution describes that variation. For independent observations from suitable population, sample mean is unbiased for μ with $\text{Var}(\bar{X})=\sigma^2/n$.

9709-37-C1

QUESTION + ANSWER

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6.5 Hypothesis tests

What is the central idea in Hypothesis tests?

State H_0 and H_1 with parameter and direction before calculating a test statistic.

9709-38-C1

QUESTION + ANSWER

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6.5 Hypothesis tests

A right-tail z test yields $p=0.032$. State decisions at 5% and 1% and explain what a 5% significance level controls.

Reject H_0 at 5%, fail to reject at 1%; 5% is maximum long-run type I error probability under H_0 .

9709-38-P

QUESTION + ANSWER